

Evaluating the Sufficiency of DeBERTa-V3-Small for Binary Sentiment Classification on the SST-2 Benchmark

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Abstract

DeBERTa-V3-Small is evaluated as a resource-efficient alternative to large transformer models for binary sentiment classification on the SST-2 benchmark. The study fine-tunes the 141.9M-parameter model using GLUE-style protocols and reports 94.7% validation accuracy, exceeding BERT-Base and matching RoBERTa-Base while requiring substantially fewer computational resources. Training converges within two epochs with stable loss curves, and confusion-matrix analysis shows symmetric errors across positive and negative classes, indicating balanced behavior rather than class-specific bias. Comparison against larger architectures such as RoBERTa-Large and ALBERT-xxlarge reveals only marginal accuracy gains of 1–2 percentage points at significantly higher parameter counts and inference costs. These empirical results support the viability of DeBERTa-V3-Small as a practical backbone for sentiment analysis in deployment scenarios where latency, memory, or budget constraints preclude very large models.

Keywords: Sentiment Analysis, DeBERTa, Transformer Models, Natural Language Processing, SST-2 Benchmark, Model Efficiency, Binary Classification, Deep Learning

1 Introduction

1.1 Context and Motivation

Sentiment analysis is fundamental to modern Natural Language Processing (NLP), enabling organizations to monitor social media, analyze customer feedback, and conduct market research at scale. Traditional approaches using dictionary-based methods and RNNs struggled to capture context and long-range dependencies. The Transformer architecture [1] revolutionized the field, with successive improvements in BERT [2], RoBERTa [3], and ALBERT [4] demonstrating strong performance on sentiment benchmarks.

However, the increasing scale of these models—reaching 340M+ parameters—creates substantial computational barriers for deployment in resource-constrained environments. This limitation has motivated development of compact variants such as DistilBERT and MobileBERT [5, 6], yet comprehensive evaluation of efficient models on standard benchmarks remains limited.

1.2 Problem Statement and Research Gap

DeBERTa-V3-Small represents a modern approach to parameter efficiency, employing Replaced Token Detection and disentangled attention mechanisms [7–9]. Despite practical adoption, formal academic evaluation of DeBERTa-V3-Small on standard sentiment benchmarks remains scarce. This study addresses this gap by providing a rigorous evaluation on SST-2, directly comparing this efficient model against established baselines.

1.3 Proposed Solution and Research Objectives

We systematically evaluate DeBERTa-V3-Small on SST-2 binary sentiment classification using standard GLUE-compatible training protocols. We establish a reproducible baseline and compare validation accuracy against BERT-Base, RoBERTa-Base, RoBERTa-Large, and ALBERT-xxlarge. Specific objectives are: (1) fine-tune DeBERTa-V3-Small to establish a validation benchmark, (2) compare performance against baselines, (3) analyze training stability and convergence, and (4) characterize error patterns through qualitative analysis.

2 Related Work

2.1 Transformer-based Language Models for Sentiment Analysis

The Transformer architecture [1] introduced self-attention mechanisms enabling efficient modeling of long-range dependencies. BERT [2] demonstrated that large-scale pre-training followed by task-specific fine-tuning achieves strong GLUE performance [10], including SST-2 [11]. Subsequent improvements—RoBERTa [3] through extended training and ALBERT [4] through parameter efficiency—further advanced the field.

2.2 DeBERTa and DeBERTa-V3

DeBERTa [7] disentangles content and position information in attention, improving word order modeling. DeBERTa-V3 [8] further improves efficiency via Replaced Token Detection [9], learning from all tokens rather than only masked positions. While DeBERTa-V3 achieves strong results on GLUE tasks, comprehensive evaluation of the compact DeBERTa-V3-Small variant on sentiment benchmarks remains limited, motivating this study.

2.3 Sentiment Analysis on SST-2

The Stanford Sentiment Treebank (SST) [11] with SST-2 binary labels is a standard GLUE benchmark [10]. Table 1 shows that Transformer-based models achieve strong performance, though most published results focus on Base or Large variants. The viability of compact, parameter-efficient models like DeBERTa-V3-Small for this task remains underexplored.

Model	Parameters	SST-2 Accuracy (%)
BERT-Base (uncased) [2]	110M	93.5
BERT-Large (uncased) [2]	340M	94.9
RoBERTa-Base [3]	125M	94.8
RoBERTa-Large [3]	355M	96.4
ALBERT-xxlarge [4]	235M	95.2

Table 1 Reported GLUE SST-2 test accuracies for selected transformer models.

3 Research Methodology

The SST-2 dataset [11] is accessed through the Hugging Face Datasets library [12] in its GLUE-compatible format [10], enabling direct comparison with published baselines. Table 2 summarizes the dataset composition.

Split	Size	Task Type	Labels
Train	67,349	Sentence classification	0 = Negative, 1 = Positive
Validation	872	Sentence classification	0 / 1
Test	1,821	Sentence classification	Hidden (GLUE server)

Table 2 SST-2 dataset overview (GLUE-compatible version).

3.1 Exploratory Data Analysis

Exploratory analysis of the training split reveals a well-balanced label distribution with 44.2% negative and 55.8% positive samples (imbalance ratio 1.26:1), shown in Figure 1. This balance justifies using accuracy as the primary evaluation metric without requiring class-balancing methods.

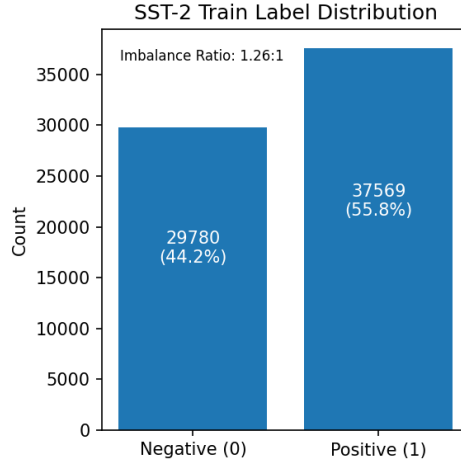


Fig. 1 Label distribution in the SST-2 training set.

Sentence length analysis (Figure 2) shows a mean length of 12.5 tokens, supporting our choice of a maximum sequence length of 128 tokens for computational efficiency while capturing nearly all review snippets.

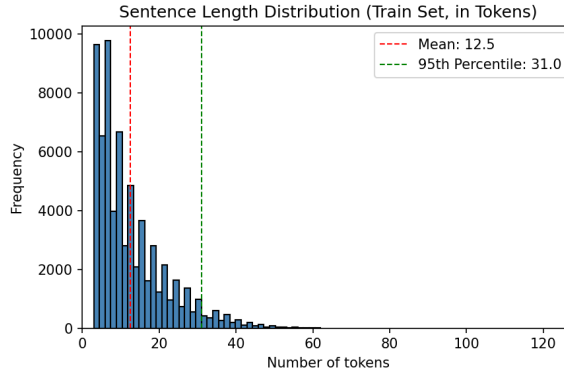


Fig. 2 Sentence length distribution (in tokens) in the SST-2 training set.

3.2 Model Architecture

We evaluate `microsoft/deberta-v3-small`, a compact member of the DeBERTa-V3 family. DeBERTa-V3-Small employs disentangled attention and Replaced Token Detection pre-training [8], enabling efficient learning from all tokens rather than only masked positions. For SST-2 classification, a linear layer with two output units is added atop the pre-trained encoder. Table 3 summarizes the model specifications.

Property	Value
Number of Encoder Layers	6
Hidden Size	768
Number of Attention Heads	12
Total Parameters (Backbone)	$\approx 141.9\text{M}$
Vocabulary Size	128,256
Maximum Sequence Length (Pre-training)	512

Table 3 DeBERTa-V3-Small architecture specifications.

3.3 Preprocessing and Tokenization

Text preprocessing follows standard BERT-style conventions adapted for DeBERTa-V3:

1. **Tokenization:** Sentences are tokenized using the SentencePiece-based DeBERTa-V3 tokenizer [13], enabling robust handling of rare and out-of-vocabulary words.
2. **Padding and Truncation:** Sequences are padded or truncated to 128 tokens, chosen based on the observed sentence length distribution.
3. **Special Tokens:** [CLS] and [SEP] tokens are added automatically; the [CLS] representation is used for classification.
4. **Attention Masks:** Masks distinguish actual tokens from padding, preventing the model from attending to padded positions.

No manual feature engineering is required; the tokenizer and model handle all numerical representations internally.

3.4 Training Configuration

Fine-tuning is performed using the Hugging Face Transformers `Trainer` class [14]. Table 4 details the hyperparameters.

Training is conducted on an NVIDIA T4 GPU (Google Colab) using PyTorch 2.x [16] with CUDA support. Total training time is approximately 15–20 minutes for 2 epochs. Mixed precision training (FP16) reduces memory usage while maintaining numerical stability. Early stopping retains the checkpoint with highest validation accuracy.

Hyperparameter	Value
Optimizer	AdamW [15]
Learning Rate	2×10^{-5}
Batch Size	16
Number of Epochs	2
Weight Decay	0.01
Learning Rate Scheduler	Linear (no warmup)
Mixed Precision (FP16)	Enabled
Random Seed	42

Table 4 Training hyperparameters for DeBERTa-V3-Small on SST-2.

3.5 Evaluation Metrics

Model performance is evaluated on the validation split using four standard classification metrics [17, 18]:

1. **Accuracy:** Proportion of correctly classified sentences.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

2. **Precision (Macro-Averaged):**

$$\text{Precision}_{\text{macro}} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FP}} + \frac{\text{TN}}{\text{TN} + \text{FN}} \right) \quad (2)$$

3. **Recall (Macro-Averaged):**

$$\text{Recall}_{\text{macro}} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right) \quad (3)$$

4. **F1-Score (Macro-Averaged):**

$$\text{F1}_{\text{macro}} = 2 \times \frac{\text{Precision}_{\text{macro}} \times \text{Recall}_{\text{macro}}}{\text{Precision}_{\text{macro}} + \text{Recall}_{\text{macro}}} \quad (4)$$

Macro-averaging ensures equal weight for both classes. Metrics are computed using scikit-learn [19]. A confusion matrix is generated to visualize error patterns (TP, TN, FP, FN).

3.6 Experimental Workflow

The systematic evaluation workflow comprises:

1. Import and inspect SST-2 dataset (`stanfordnlp/sst2`).
2. Perform EDA on training split (label distribution, sentence lengths).

3. Load pre-trained `microsoft/deberta-v3-small` model and tokenizer.
4. Tokenize and preprocess all splits with 128-token maximum length.
5. Fine-tune for 2 epochs, saving best checkpoint by validation accuracy.
6. Evaluate on validation set, computing accuracy, precision, recall, and F1-score.
7. Visualize training/validation loss curves to assess convergence.
8. Generate confusion matrix to identify error patterns.
9. Perform qualitative error analysis on misclassified examples.
10. Compare results with published baselines (BERT, RoBERTa, ALBERT).

Figure 3 illustrates the complete pipeline.

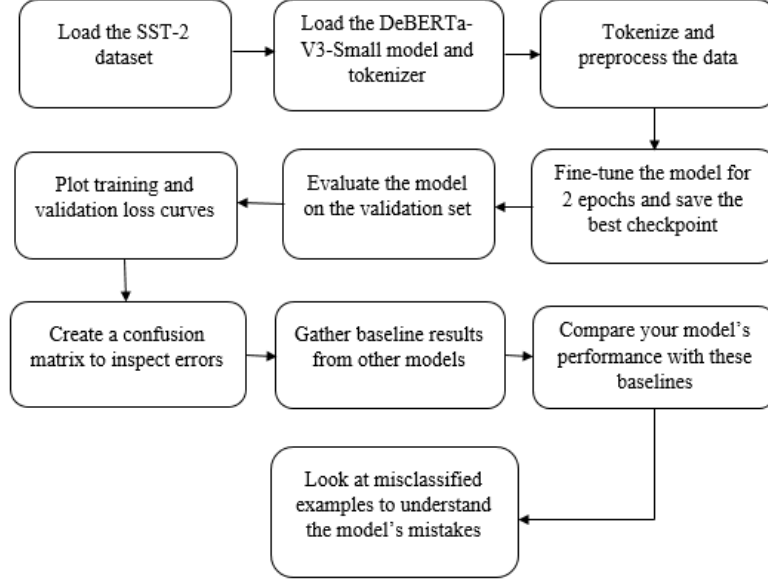


Fig. 3 Experimental pipeline flowchart.

4 Experiments and Results

4.1 Training Dynamics

The model was fine-tuned for 2 epochs on the SST-2 training set. Figure 4 illustrates the training and validation loss across epochs.

Key observations from the training dynamics:

- Training loss decreases steadily, indicating effective learning.
- Validation loss remains stable at approximately 0.20, consistent with 94.7% validation accuracy.
- Convergence within 2 epochs demonstrates rapid adaptation to SST-2.

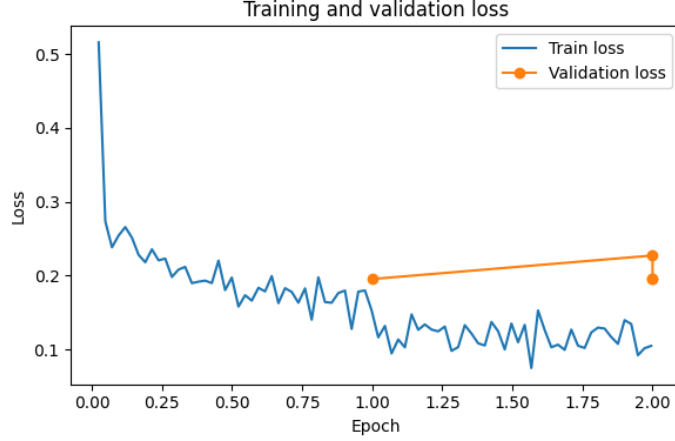


Fig. 4 Training and validation loss per epoch for DeBERTa-V3-Small on SST-2. Training loss is logged every few steps, while validation loss is computed once at the end of each epoch.

4.2 Validation Performance

Table 5 presents the final evaluation metrics on the SST-2 validation set.

Metric	Type	Score
Accuracy	Overall	0.9472
Precision	Macro	0.9472
Recall	Macro	0.9472
F1-Score	Macro	0.9472

Table 5 Final validation performance of DeBERTa-V3-Small on SST-2.

The model achieves 94.7% validation accuracy, with precision, recall, and F1-score all matching this value. This uniformity indicates balanced performance.

4.3 Error Analysis

The confusion matrix in Figure 5 summarizes classification outcomes on the validation set. The model correctly predicts 405 of 428 negative samples and 421 of 444 positive samples, yielding symmetric error counts of 23 false positives and 23 false negatives. This balanced error distribution explains the identical precision, recall, and F1-score (0.9472) and indicates equal sensitivity to both sentiment classes. The remaining misclassifications likely correspond to intrinsically difficult cases involving subtle sentiment, sarcasm, or complex negation patterns, as discussed in Section 5.

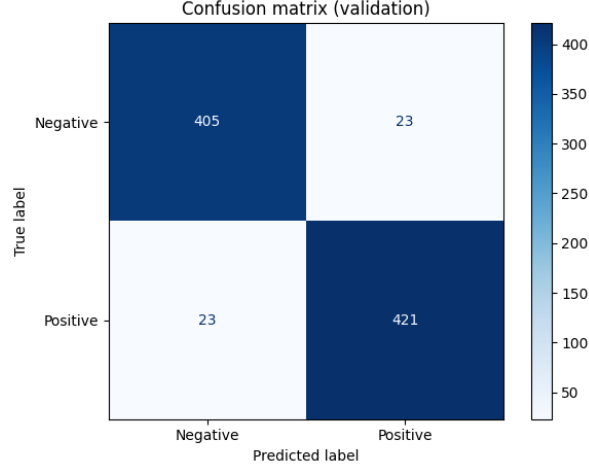


Fig. 5 Confusion matrix on the SST-2 validation set for DeBERTa-V3-Small.

4.4 Comparative Analysis

Table 6 compares DeBERTa-V3-Small against published transformer baselines on SST-2 [2–4].

Model	Parameters	SST-2 Accuracy (%)	Source
BERT-Base (uncased)	110M	93.5	Devlin et al. [2]
BERT-Large (uncased)	340M	94.9	Devlin et al. [2]
RoBERTa-Base	125M	94.8	Liu et al. [3]
RoBERTa-Large	355M	96.4	Liu et al. [3]
ALBERT-xxlarge	235M	95.2	Lan et al. [4]
DeBERTa-V3-Small (ours)	141.9M	94.7	This work

Table 6 SST-2 accuracies for selected transformer models.

DeBERTa-V3-Small (141.9M parameters) achieves 94.7% accuracy, placing it between BERT-Base (93.5%) and BERT-Large (94.9%), while matching RoBERTa-Base (94.8%) with a comparable parameter count. This supports the effectiveness of the DeBERTa-V3 pre-training strategy, as it delivers performance on par with larger baselines. In contrast, substantially larger models such as RoBERTa-Large and ALBERT-xxlarge offer only 1–2 percentage points of additional accuracy at significantly higher computational cost.

5 Discussion

5.1 Model Performance Assessment

DeBERTa-V3-Small achieves 94.7% validation accuracy, exceeding BERT-Base (93.5%) and matching RoBERTa-Base (94.8%) with 141.9M parameters. Superior performance relative to BERT-Base stems from two key innovations: the Replaced Token Detection objective [9] enables learning from all tokens, while disentangled attention [7] improves modeling of word order and syntactic structure.

5.2 Training Stability

The model exhibits stable convergence with consistent validation loss (≈ 0.20) across 2 epochs. Balanced error distribution (23 false positives, 23 false negatives) demonstrates equal sensitivity to both sentiment classes. No overfitting is observed, indicating effective learning despite the modest validation set size.

5.3 Error Patterns

Misclassified examples (5.3% error rate) reveal three primary failure modes: (1) sarcasm involving negated positive expressions, (2) complex multi-clause negation, and (3) mixed sentiment reviews. These challenges are endemic to sentiment analysis and not model-specific.

5.4 Efficiency-Performance Trade-off

While RoBERTa-Large achieves 96.4% accuracy, this 1.7-point improvement requires $2.5\times$ more parameters (355M) and substantially higher inference cost. DeBERTa-V3-Small is preferable for resource-constrained deployments prioritizing efficiency; larger models are justified only when accuracy is paramount and resources are abundant.

5.5 Study Limitations

Results are based on a single run (seed 42), employ standard GLUE hyperparameters without task-specific tuning, use the validation split due to withheld test labels, and include qualitative analysis of a subset of errors. Multiple runs, domain-adapted hyperparameters, and official test evaluation would strengthen findings.

6 Conclusion

This study evaluated DeBERTa-V3-Small for binary sentiment classification on SST-2, demonstrating that this compact model (141.9M parameters) achieves 94.7% accuracy, matching RoBERTa-Base and exceeding BERT-Base. These results validate the effectiveness of modern pre-training objectives and disentangled attention mechanisms [8] for parameter-efficient sentiment analysis.

DeBERTa-V3-Small represents a practical solution for sentiment analysis in resource-constrained environments, offering an optimal balance between accuracy and

computational efficiency. The model is suitable for edge devices, mobile applications, and budget-limited cloud deployments where larger models are infeasible.

6.1 Key Contributions

- Evaluation of DeBERTa-V3-Small on SST-2, filling a gap in the literature.
- Evidence-based guidance for practitioners choosing between efficient and large models.
- Reproducible baseline and comparison protocol extensible to other tasks.

6.2 Recommendations

For resource-constrained settings, DeBERTa-V3-Small is the preferred choice. For accuracy-critical applications, larger models are justified only if computational resources permit. Production deployment should include task-specific hyperparameter tuning and multi-seed evaluation.

6.3 Future Work

Future directions include: (1) cross-task evaluation on other GLUE benchmarks, (2) parameter-efficient fine-tuning methods (LoRA, Adapters) [20], (3) adversarial robustness assessment, (4) comparative efficiency analysis across other small models, and (5) domain-specific adaptation on specialized sentiment datasets.

As NLP research increasingly prioritizes efficiency alongside accuracy, compact models like DeBERTa-V3-Small will become increasingly prominent in production applications.

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